

Trainings ▾

General Information

Engine coolant is composed of three main components: water, glycol or antifreeze, and an additive package.

Water

Engine coolant is approximately 50 percent water by volume. Water is primarily responsible for the following:

- Removing heat from the engine and transferring it to the environment
- Dissolving the additive package
- Maintaining a coolant viscosity that can be pumped through the system at the required rate.

Glycol/Antifreeze

Antifreeze is approximately 50 percent of engine coolant by volume. Below 40 percent concentration, coolant is susceptible to freezing. Above 60 percent concentration, coolant is susceptible to additive precipitation and silicate instability.

Antifreeze is used to assist with the following:

- Lowers freezing point
- Raises boiling point
- Lowers vapor pressure of mixture.

Antifreeze consists of Ethylene Glycol (EG) or Propylene Glycol (PG). Ethylene glycol and propylene glycol are similar, but the following should be noted:

- Ethylene glycol is a slightly more efficient heat transfer fluid under convective heat transfer conditions
- Ethylene glycol products are traditionally used in diesel engine antifreeze
- Ethylene glycol should be utilized in arctic climates due to lower viscosity
- Propylene glycol is slightly more viscous
- Propylene glycol is suited for applications that require less noxious products.

Additives

The additive package typically makes up 1 to 10 percent of the coolant. Additives are used in coolant for liner pitting protection, corrosion protection, pH buffering, and scale and deposit protection.

Coolants are classified based on the type of additives they contain for liner pitting and corrosion protection. There are three broad categories a coolant could be classified into, based on the additive package.

- **Conventional** – Additive package consists of predominantly inorganic compounds; nitrite, molybdate, phosphate, borate, etc.
- **Hybrid** – Additive package consists of a mixture of inorganic and organic acid salt components. Hybrid coolants may be advertised as HOAT (Hybrid Organic Acid Technology) or NOAT (Nitrited Organic Acid Technology).
- **Extended Life/Organic Acid Technology (OAT)** - The additive package consists mostly of organic acids; sebacate, benzoate, adipate, etc. OAT coolants will generally **not** contain nitrite, amines, borate, phosphate, or silicate. OAT additives have a much longer service life than inorganic additives.